PURPOSE-LED PUBLISHING™

PAPER • **OPEN ACCESS**

2D graded tensegrity-inspired metamaterials with hexagonal cells fabricated via additive manufacturing

To cite this article: Wojciech Gilewski *et al* 2025 *2D Mater.* **12** 045026

View the article online for updates and enhancements.

You may also like

- Experiments and parametric studies on 3D metallic auxetic metamaterials with tuneable mechanical properties
Xin Ren, Jianhu Shen, Arash Ghaedizadeh et al.
- High-stiffness mechanical metamaterials based on hybrid interpenetrating structures: design and research
Zhongchun Wen, Feng Guan, Hongzhi Xie et al.
- A novel chiral metamaterial with multistability and programmable stiffness
Jia-Xin Wang, Qing-Sheng Yang, Yu-Ling Wei et al.

This content was downloaded from IP address 103.232.241.248 on 10/11/2026 at 12:51

2D Materials

Check for updates

PAPER

2D graded tensegrity-inspired metamaterials with hexagonal cells fabricated via additive manufacturing

Wojciech Gilewski[1], **Adam Zawadzki**[2], **Marta Thammavong**[1] and **Anna Al Sabouni-Zawadzka**[1,*]

[1] Faculty of Civil Engineering, Warsaw University of Technology, Warsaw, Poland
[2] Faculty of Automotive and Construction Machinery, Warsaw University of Technology, Warsaw, Poland
* Author to whom any correspondence should be addressed.

E-mail: anna.zawadzka@pw.edu.pl

Keywords: tensegrity-inspired lattices, 2D mechanical metamaterials, graded structures, additive manufacturing, hexagonal cells

OPEN ACCESS

RECEIVED
22 August 2025

REVISED
16 October 2025

ACCEPTED FOR PUBLICATION
22 October 2025

PUBLISHED
31 October 2025

Original Content from this work may be used under the terms of the Creative Commons Attribution 4.0 licence.

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.

Abstract

This paper presents a computational and experimental study of two-dimensional (2D) tensegrity-inspired metamaterials with graded mechanical properties, fabricated via additive manufacturing (AM). Hexagonal tensegrity modules with different topologies and proportions were analysed using a continuum model of tensegrity with closed-form expressions for equivalent elastic properties. The results reveal the occurrence of extremal mechanical properties, particularly in the rotated configuration of the hexagonal module, enabling significant tailoring of stiffness and deformation behaviour. Based on these findings, several lattice configurations with uniform and graded properties were designed and produced in a medium scale using MultiJet Printing (MJP) technology, taking advantage of its precision and ease of post-processing. Mechanical testing of additively manufactured samples under both uniformly distributed and concentrated compression loads demonstrated that gradation of the module geometry and layer arrangement allows control over stiffness, load-bearing capacity, and the extent of mechanically stable regions. The study establishes a computational—experimental framework for the preliminary evaluation of 2D graded tensegrity-inspired lattices and highlights the potential of AM to realise programmable mechanical metamaterials. The results in the scale considered are promising and motivate further research into the static and dynamic properties of such systems for advanced engineering applications.

1. Introduction

The field of mechanical metamaterials has recently witnessed breakthroughs that have enabled unprecedented functionalities. Researchers have engineered materials that defy conventional mechanical behaviour, exhibiting features such as programmable response, auxetic properties, vanishing shear modulus, negative compressibility, etc [1–3]. These advances have been driven not only by innovative geometric design but also by the exploration of new classes of base materials that provide exceptional performance at the micro- and nanoscale.

Among them, two-dimensional (2D) materials, such as graphene, have attracted intense research interest due to their unique mechanical properties [4–6]. 2D materials offer exceptional stiffness, strength, and flexibility, making them ideal platforms for architected mechanical systems. Their properties can be tuned through strain engineering, defect design, and heterostructure assembly, allowing control at the material and structural level [5]. Advances in the synthesis and testing of 2D materials [6, 7] have enabled their applications in flexible electronics, sensors, nanomechanical devices, composites, and biomedical systems.

Combining ideas from both advanced materials and metamaterial concepts [8], 2D mechanical metamaterials have emerged as programmable sheets capable of extreme deformation, on-demand stiffness tuning, and precise wave control, among others. Different 2D lattice layouts have been studied in search of unique mechanical characteristics such as vibration suppression, energy absorption, and tailored thermal expansion [9]. Bimaterial serpentine lattices with micrometre features have been developed, offering highly tunable thermal expansion and rapid and reversible shape changes [10]. In 2D multistable mechanical metamaterials, colliding wave pulses have been used to trigger and control phase

© 2025 The Author(s). Published by IOP Publishing Ltd

changes that propagate through the structure [11]. The introduction of topological defects in additively manufactured 2D honeycomb metamaterials has also been found to significantly influence their mechanical properties and failure mechanisms [12].

Several studies have also focused on designing 2D mechanical metamaterials with auxetic behaviour. Negative thermal expansion designs have been used to create structures with tunable Poisson's ratios [13], while lattices with peanut-shaped holes have been proposed to enhance auxetic performance [14]. Machine learning methods combined with molecular dynamics simulations have enabled the design of structures with tunable deformation-dependent Poisson's ratios [15]. In addition, chiral metamaterials based on monohedral pentagonal tessellations have been shown to achieve Poisson's ratios ranging from -1 to $+1$ with highly tunable stiffness [16].

An interesting approach to designing mechanical metamaterials, well established in three-dimensional (3D) structures but not yet explored in 2D systems, is to use tensegrity-based cells assembled into larger lattices. Tensegrity structures are pin-jointed systems composed of compression members (struts) interconnected by tension members (cables) [17]. Their unique properties arise from infinitesimal mechanisms which are stabilised via self-equilibrated systems of normal forces (self-stress states) [18, 19]. The occurrence of self-stress states makes it possible to create systems with highly non-linear response and tunable mechanical behaviour [20–22].

Additive manufacturing (AM) has enabled the production of tensegrity-based metamaterials at various scales. Early works [23–25] have focused on fabricating pure tensegrity systems, that is, with pin-jointed connections between struts and cables. However, more recent studies have used lattices with rigid connections, usually referred to as *tensegrity-like* or *tensegrity-inspired* systems. As several studies have shown [26–28], such lattices can still exhibit mechanical behaviour similar to that of pure tensegrity structures.

Based on this concept, bistable lattices with a tensegrity architecture and nanoscale features have been developed [29], and tensegrity-inspired lattices with tunable bandgaps [30] and dynamic attenuation [31] have been introduced. Lattice chains exhibiting multistable behaviour have also been investigated [32]. More recently, the stability of tensegrity-inspired units has been examined [33], and modular tensegrity-like lattices with auxetic properties and geometrically non-linear response have been developed [22].

The aim of the present study is to model 2D tensegrity-inspired lattices with graded mechanical properties to enable control over stiffness, load-bearing capacity, and the extent of mechanically stable regions understood as areas that are practically not subject to deformation. To the best of the authors' knowledge, there is a gap in this area in literature.

The authors have previously investigated 2D tensegrity systems, focusing on their correct theoretical classification and definition using matrix analyses [34]. The present work extends the concept toward tensegrity-inspired architected lattices, where the tensegrity principle is used as a design motif to develop graded metamaterials with controllable mechanical properties. The computational approach adopted here is based on a continuum model proposed in [35] and described in detail in [36]. In contrast to those studies, which focused primarily on formulating and presenting the model, the present paper applies this continuum-based approach to the analysis of 2D lattice structures.

The computational framework developed here is intended for preliminary evaluation of the mechanical behaviour of additively manufactured systems and for providing recommendations on the arrangement of cells and layers. As the proposed computational models are not intended to replace or replicate real metamaterials, no calibration or validation against the experimental results was performed. To inform the modelling approach, preliminary experimental investigations were carried out on additively manufactured tensegrity-inspired lattices to qualitatively assess their mechanical behaviour. Several regular and graded lattices, each composed of a 10×6 array of hexagonal modules, were examined under both uniform compression and concentrated loading, and the influence of the introduced gradation was analysed.

2. Selection of the hexagonal tensegrity cell

The focus of this work is on tensegrity modules based on regular or irregular hexagons [34]. This class of modules offers great possibilities for creating metamaterial structures with graded mechanical properties.

The pure tensegrity module is a pin-jointed cable-strut structure with two possible basic topologies within hexagonal cells, as shown in figures 1(a) and (c). It should be noted that the struts, shown as thick lines, do not connect with each other inside the hexagons (they pass each other), and all nodes are pin-jointed. It is possible to rotate the base cells by any angle, i.e. the Hexa-1 module (figure 1(a)) rotated by 90° is shown in figure 1(b), and by 60°—in figure 1(d).

First, let us analyse the continuum model of basic cells introduced in [35] and further developed in [36]. The model leads to the elasticity matrix $\mathbf{E}$ in terms of the plane stress linear elasticity theory [37], which describes the mechanical properties of the cell/module.

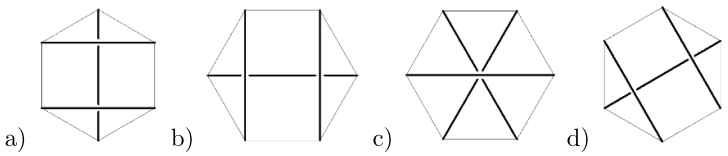


Figure 1. Hexagonal cells: (a) Hexa-1 module; (b) Hexa-2 module (Hexa-1 rotated by 90°); (c) Hexa-3 module; (d) Hexa-4 module (Hexa-1 rotated by 60°).

The substitute elasticity matrices of the analysed modules, their eigenvalues, and eigenvectors are shown below. As the Hexa-2 module (figure 1(b)) is the same as Hexa-1, only rotated by 90°, the mechanical properties of these modules are analogous if we change the direction of observation. For the Hexa-1 module (figure 1(a)), we obtain the following elasticity matrix and its eigenvalues and eigenvectors:

$$\mathbf{E}_{\text{Hexa-1}} = \frac{EA}{8ah}\begin{bmatrix} 8+3\sqrt{3}k+2\sigma & \sqrt{3}k-2\sigma & 0 \\ \sqrt{3}k-2\sigma & \frac{1}{3}\left(8\sqrt{3}+9\sqrt{3}k+6\sigma\right) & 0 \\ 0 & 0 & \sqrt{3}k-2\sigma \end{bmatrix}, \tag{1}$$

$$\begin{aligned} \lambda_{1,\text{Hexa-1}} &= \frac{EA}{8ah}\left(\sqrt{3}k-2\sigma\right), \\ \lambda_{2,\text{Hexa-1}} &= \frac{EA}{24ah}\left(12+4\sqrt{3}+9\sqrt{3}k+6\sigma-\sqrt{3}\sqrt{64-32\sqrt{3}+9k^2-12\sqrt{3}k\sigma+12\sigma^2}\right), \\ \lambda_{3,\text{Hexa-1}} &= \frac{EA}{24ah}\left(12+4\sqrt{3}+9\sqrt{3}k+6\sigma+\sqrt{3}\sqrt{64-32\sqrt{3}+9k^2-12\sqrt{3}k\sigma+12\sigma^2}\right), \end{aligned} \tag{2}$$

with the parameters defined as:

$$k = \frac{(EA)_{\text{cable}}}{(EA)_{\text{strut}}}, \quad (EA)_{\text{strut}} = EA, \quad \sigma = \frac{S_0}{EA}, \tag{3}$$

where:

k—cable-to-strut stiffness ratio,

σ—level of self-stress,

E—Young's modulus,

A—cross-sectional area of the member,

a—edge length of the regular hexagon,

h—thickness of the 2D elastic area equivalent to the module,

S_0—multiplier of self-equilibrated system of normal forces.

For the Hexa-2 module (figure 1(b)), elements 11 and 22 in the elasticity matrix should be swapped, but the eigenvalues are identical. It can be proved that the second and third eigenvalues of both matrices are always positive. However, the first eigenvalue may be zero when

$$\sigma_{\text{soft,Hexa-1,2}} = \frac{\sqrt{3}k}{2}, \tag{4}$$

which is also the boundary line of the positive definiteness of the elasticity matrix.

For the boundary line, in both cases (Hexa-1 and Hexa-2), we are dealing with a zero form of energy represented by the eigenvector

$$\mathbf{w}_{\text{soft,Hexa-1,2}} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T, \tag{5}$$

which describes pure shear deformation. The results indicate extremal uni-mode properties, as described in [38]. Extremal materials, first introduced in [39], are extremely stiff materials in some deformation modes while highly compliant in others, often showing unusual mechanical behaviours, such as a negative Poisson's ratio, vanishing shear modulus, negative compressibility, or negative stiffness [1, 40]. The uni-mode materials are characterised by one compliant mode of deformation, which is further referred to as the soft deformation mode.

For the Hexa-3 module (figure 1(c)):

$$\mathbf{E}_{\text{Hexa-3}} = \frac{\sqrt{3}EA\,(1+k)}{8ah}\begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{6}$$

$$\lambda_{1,\text{Hexa-3}} = 2\lambda_{2,\text{Hexa-3}} = 4\lambda_{3,\text{Hexa-3}} = \frac{EA\sqrt{3}\,(1+k)}{2ah}. \tag{7}$$

The coefficients of the elasticity matrix do not depend on the level of self-stress σ, and all the eigenvalues are positive. The module under consideration does not exhibit extremal properties.

Another possible topology of the Hexa-1 module is a configuration rotated by 60° (figure 1(d)), referred to as Hexa-4. In addition, this module can be easily connected to other hexagons to form a variety of lattices. The elasticity matrix now has the form:

$$\mathbf{E}_{\text{Hexa-4}} = \frac{EA}{16ah}\begin{bmatrix} 1+3\sqrt{3}+6\sqrt{3}k-2\sigma & 3+\sqrt{3}+2\sqrt{3}k+2\sigma & -3+\sqrt{3}-2\sqrt{3}\sigma \\ 3+\sqrt{3}+2\sqrt{3}k+2\sigma & 9+\frac{1}{3}\sqrt{3}+6\sqrt{3}k-2\sigma & -1+3\sqrt{3}+2\sqrt{3}\sigma \\ -3+\sqrt{3}-2\sqrt{3}\sigma & -1+3\sqrt{3}+2\sqrt{3}\sigma & 3+\sqrt{3}+2\sqrt{3}k+2\sigma \end{bmatrix}. \tag{8}$$

The eigenvalues of the Hexa-4 module's elasticity matrix are the same as those of the non-rotated module, as is the relationship that describes soft extremal values. However, an important difference is the different mode of soft deformation described by the vector

$$\mathbf{w}_{\text{soft,Hexa-4}} = \begin{bmatrix} -0.547723 & 0.547723 & 0.632456 \end{bmatrix}^{\text{T}}. \tag{9}$$

This mode corresponds to compression in the vertical direction with tension in the horizontal direction and shear deformation.

The analysis presented above indicates that the Hexa-1, Hexa-2, and Hexa-4 modules are richer in terms of possible unique mechanical properties than the Hexa-3 module. Therefore, these modules will be the subject of further analysis. This investigation is supported by an open-access dataset that provides universal Mathematica codes for the continuum analysis of 2D tensegrity modules with arbitrary geometry [41]. In the next part of this paper, a comparative analysis is presented showing how the properties of the modules change with different geometric proportions.

3. Evaluation of mechanical properties of hexagonal cells

Based on the elasticity matrices presented in section 2, the formulas for equivalent technical coefficients can be determined and analysed according to [36]. The elasticity matrices of the Hexa-1 and Hexa-2 modules are orthotropic. For the non-rotated Hexa-1 module, we obtain the following Young's moduli, shear moduli, and Poisson's ratios (1 represents the horizontal direction and 2 corresponds to the vertical one):

$$\begin{aligned} E_{1,\text{Hexa-1}} &= \frac{EA}{ah}\cdot\frac{8\sqrt{3}+9k^2+2\left(3+\sqrt{3}\right)\sigma+k\left(9+9\sqrt{3}+6\sqrt{3}\sigma\right)}{8\sqrt{3}+9\sqrt{3}k+6\sigma}, \\ E_{2,\text{Hexa-1}} &= \frac{EA}{3ah}\cdot\frac{8\sqrt{3}+9k^2+2\left(3+\sqrt{3}\right)\sigma+k\left(9+9\sqrt{3}+6\sqrt{3}\sigma\right)}{8+3\sqrt{3}k+2\sigma}, \\ G_{12,\text{Hexa-1}} &= \frac{EA}{8ah}\left(\sqrt{3}k-2\sigma\right), \\ \nu_{12,\text{Hexa-1}} &= \frac{3\sqrt{3}k-6\sigma}{8\sqrt{3}+9\sqrt{3}k+6\sigma}, \\ \nu_{21,\text{Hexa-1}} &= \frac{\sqrt{3}k-2\sigma}{8+3\sqrt{3}k+2\sigma}. \end{aligned} \tag{10}$$

Changing directions 1 and 2 allows us to specify the parameters of the Hexa-2 module.

The elasticity matrix of the Hexa-4 module is anisotropic, so there are additional material coefficients, which do not occur in other orthotropic modules. For the Hexa-4 module, rotated by 60°, we obtain:

$$E_{1,\text{Hexa-4}} = \frac{EA}{ah}\cdot\frac{9\left(3+\sqrt{3}\right)k^2+9\sqrt{3}k^3-4\sigma\left[4\sqrt{3}+\left(3+\sqrt{3}\right)\sigma\right]-12k\left(-2+\sigma+\sqrt{3}\sigma+\sqrt{3}\sigma^2\right)}{27k^2+3k\left(5+9\sqrt{3}+2\sqrt{3}\sigma\right)+2\left[6\sqrt{3}+\left(-9+\sqrt{3}\right)\sigma-6\sigma^2\right]},$$

$$E_{2,\text{Hexa-4}} = \frac{EA}{3ah}\cdot\frac{9\left(3+\sqrt{3}\right)k^2+9\sqrt{3}k^3-4\sigma\left[4\sqrt{3}+\left(3+\sqrt{3}\right)\sigma\right]-12k\left(-2+\sigma+\sqrt{3}\sigma+\sqrt{3}\sigma^2\right)}{4\sqrt{3}+9k^2-2\left(-1+\sqrt{3}\right)\sigma-4\sigma^2+k\left(9+5\sqrt{3}+2\sqrt{3}\sigma\right)},$$

$$G_{12,\text{Hexa-4}} = -\frac{EA}{4ah}\cdot\frac{9\left(3+\sqrt{3}\right)k^2+9\sqrt{3}k^3-4\sigma\left[4\sqrt{3}+\left(3+\sqrt{3}\right)\sigma\right]-12k\left(-2+\sigma+\sqrt{3}\sigma+\sqrt{3}\sigma^2\right)}{-4\sqrt{3}-9\left(1+\sqrt{3}\right)k-18k^2+2\left(3+\sqrt{3}\right)\sigma+6\sqrt{3}k\sigma},$$

$$\nu_{12,\text{Hexa-4}} = \frac{3\left[4\sqrt{3}+3k^2+2\left(3+\sqrt{3}\right)\sigma+4\sigma^2+k\left(3+3\sqrt{3}+2\sqrt{3}\sigma\right)\right]}{27k^2+3k\left(5+9\sqrt{3}+2\sqrt{3}\sigma\right)+2\left[6\sqrt{3}+\left(-9+\sqrt{3}\right)\sigma-6\sigma^2\right]},$$

$$\nu_{21,\text{Hexa-4}} = \frac{4\sqrt{3}+3k^2+2\left(3+\sqrt{3}\right)\sigma+4\sigma^2+k\left(3+3\sqrt{3}+2\sqrt{3}\sigma\right)}{4\sqrt{3}+9k^2-2\left(-1+\sqrt{3}\right)\sigma-4\sigma^2+k\left(9+5\sqrt{3}+2\sqrt{3}\sigma\right)},$$

$$\kappa_{11,\text{Hexa-4}} = \frac{12\left[2+2\sqrt{3}\sigma+k\left(\sqrt{3}+3\sigma\right)\right]}{27k^2+3k\left(5+9\sqrt{3}+2\sqrt{3}\sigma\right)+2\left[6\sqrt{3}+\left(-9+\sqrt{3}\right)\sigma-6\sigma^2\right]},$$

$$\kappa_{21,\text{Hexa-4}} = -\frac{4\left(2+3k\right)\left(1+\sigma\right)}{4\sqrt{3}+9k^2-2\left(-1+\sqrt{3}\right)\sigma-4\sigma^2+k\left(9+5\sqrt{3}+2\sqrt{3}\sigma\right)},$$

$$\lambda_{11,\text{Hexa-4}} = -\frac{6+6\sqrt{3}\sigma+3k\left(\sqrt{3}+3\sigma\right)}{-4\sqrt{3}-9\left(1+\sqrt{3}\right)k-18k^2+2\left(3+\sqrt{3}\right)\sigma+6\sqrt{3}k\sigma},$$

$$\lambda_{12,\text{Hexa-4}} = -\frac{3\left(2+3k\right)\left(1+\sigma\right)}{4\sqrt{3}+18k^2-2\left(3+\sqrt{3}\right)\sigma+k\left(9+9\sqrt{3}-6\sqrt{3}\sigma\right)}. \tag{11}$$

For the three Hexa topologies 1, 2, and 4, the invariants of the tensor corresponding to the elasticity matrices are the same and have the form:

$$\begin{aligned} I_1 &= \frac{EA\left(3+\sqrt{3}+3\sqrt{3}k\right)}{3ah}, \\ I_2 &= \frac{E^2A^2\left[30\left(1+\sqrt{3}\right)k+45k^2-4\left(-4\sqrt{3}+\left(3+\sqrt{3}\right)\sigma+3\sigma^2\right)\right]}{48a^2h^2}, \\ I_3 &= \frac{E^3A^3\left(\sqrt{3}k-2\sigma\right)\left[8\sqrt{3}+9k^2+2\left(3+\sqrt{3}\right)\sigma+k\left(9+9\sqrt{3}+6\sqrt{3}\sigma\right)\right]}{96a^3h^3}. \end{aligned} \tag{12}$$

Figure 2 shows a graphical comparison of four technical coefficients of Hexa-1, Hexa-2, and Hexa-4 for parameters $0 < k < 1$ and $0 < \sigma < 0.2$. The graph for the coefficient ν_{21} qualitatively corresponds to the graph for ν_{12}.

The results highlight the differences in mechanical properties between the three configurations analysed. For Young's modulus E_1 (figure 2(a)), Hexa-1 exhibits the highest stiffness in the horizontal direction, followed by Hexa-2 and Hexa-4, with the values remaining relatively insensitive to changes in k and σ. In contrast, for Young's modulus E_2 (figure 2(b)), Hexa-2 shows the highest values within the range of parameters considered, while Hexa-1 and Hexa-4 exhibit lower stiffness in the vertical direction. The grey area in figures 2(c) and partly 2(d) concerns the range of k and σ for which the elasticity matrix is not positive definite (see equation (4)). We do not consider this part of the graphs in this paper. Shear modulus G_{12} (figure 2(c)) increases non-linearly with k, with Hexa-4 indicating enhanced shear resistance compared to the other two configurations. Finally, for Poisson's ratio ν_{12} (figure 2(d)), Hexa-4 achieves the highest values, especially at low k and high σ, while Hexa-1 and Hexa-2 remain comparatively lower.

The second group of analyses concerns the parametrisation of the non-rotated Hexa-1 module depending on its height. Changes in module height are considered at a constant width of the module and a constant spacing ratio of the struts. The parameter β describes the height-to-width ratio of the module, with $\beta = 1$ indicating the regular Hexa-1 module, as shown in figure 3.

Figure 4 shows a graphical comparison of four technical coefficients of the Hexa-1 module depending on its height ratio β and the parameter k, for $\sigma = 0.01$. The value of σ adopted for the analyses is justified by the fact that the mechanical properties of the Hexa-1 module are only slightly sensitive to changes in this parameter in the considered range. The graph for the coefficient ν_{21} qualitatively corresponds to the graph for ν_{12}.

The results for Hexa-1 with varying geometrical proportions reveal distinct trends in its mechanical response. For Young's modulus E_1 (figure 4(a)), very high values are obtained at low β (tending towards infinity for $\beta \to 0$), which rapidly decrease and stabilise as the module height increases, while the effect of k remains negligible. In contrast, Young's modulus E_2 (figure 4(b)) shows a gradual increase with both parameters. Shear modulus G_{12} (figure 4(c)) remains low across most of the parameter space, but exhibits a sharp rise at high k combined with low β, indicating enhanced shear resistance under these conditions. Local extremes of the values G_{12} and ν_{12} (figures 4(c) and (d)) occur for the values of β close to 1, i.e. for a module inscribed in a regular hexagon. Poisson's ratio ν_{12} (figure 4(d)) demonstrates a similar dependency to shear modulus, staying relatively small for most values of k and β, but increasing steeply in the same region of high k and low β, suggesting strong coupling between lateral and longitudinal deformations in this regime.

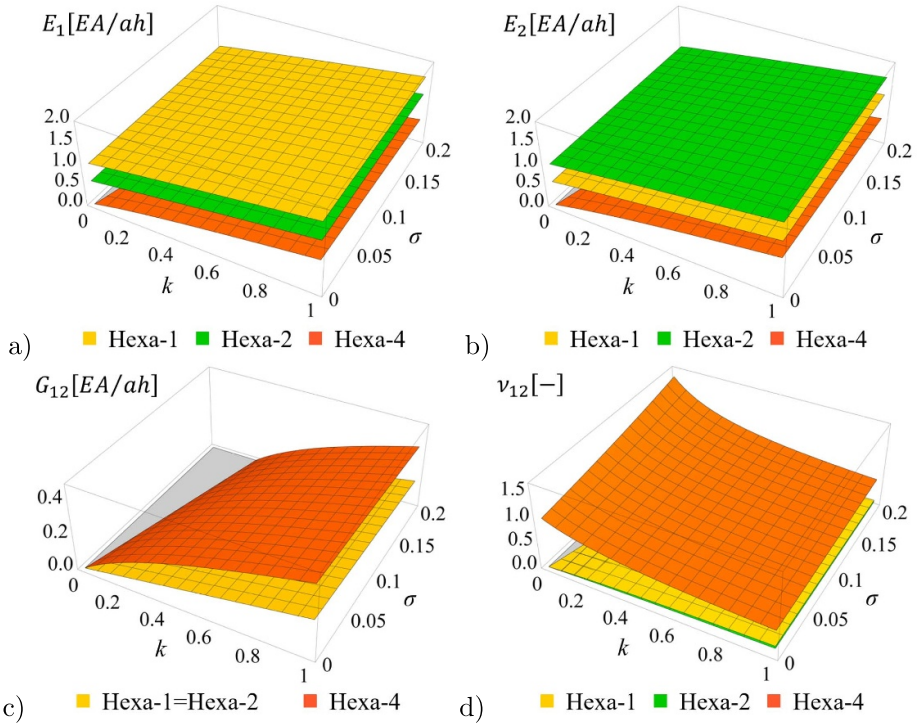


Figure 2. Graphical comparison of technical coefficients for Hexa-1, Hexa-2, and Hexa-4: (a) Young's modulus E_1; (b) Young's modulus E_2; (c) shear modulus G_{12}; (d) Poisson's ratio ν_{12}.

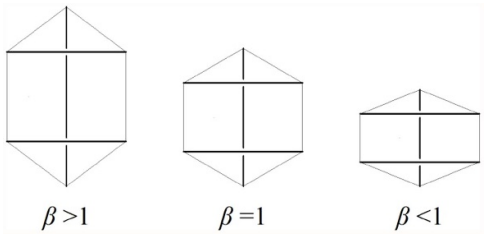


Figure 3. Parametrisation of the Hexa-1 module depending on its height.

4. Recommendations for modelling lattices

The continuum analysis presented in Sections 2 and 3 allowed us to propose six variants of the hexagonal module for use in tensegrity lattices with uniform and graded mechanical properties. It should be noted that a direct quantitative comparison between experimental measurements and theoretical estimations using a continuum model is not possible. The estimation is intended to compare the mechanical properties of cells with different characteristics in order to propose a gradation of these properties.

The geometric and mechanical properties (technical coefficients) of these modules are summarised in table 1. Numerical values are given for the geometric parameters assumed in additively manufactured samples. The dimensions of the samples and diameters of the elements were dictated by the technical limitations of the applied manufacturing technique, as well as by the need to accommodate six layers of cells to ensure the implementation of different types of gradation within the lattice, and the need to allow visual observation of deformations during experiments. Since the value of σ has a relatively minor influence on the numerical values of the elasticity coefficients and, in particular, does not significantly affect their relative proportions (see figure 2), it was arbitrarily assumed to be 0.01. The choice of $\sigma = 0.01$ also helps to prevent the potential local instability of the struts, whose effect was discussed in [33]. A comparative multiplier E is introduced to ensure that the results are valid for samples manufactured from resins with different Young's moduli.

Based on the coefficients of the inverse elasticity matrix, it is possible to assess the deformation capacity of a given system:

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & \frac{\lambda_{11}}{G_{12}} \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & \frac{\lambda_{12}}{G_{12}} \\ \frac{\kappa_{11}}{E_1} & \frac{\kappa_{21}}{E_2} & \frac{1}{G_{12}} \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix}. \quad (13)$$

Individual columns of the matrix can be interpreted as a description of deformations resulting, for example, from uniform stress states. In a testing machine, we achieve a state close to $\sigma_y = const$. The deformations are determined by Young's modulus E_2 and the coefficients ν_{12}, κ_{21}. It is also possible to

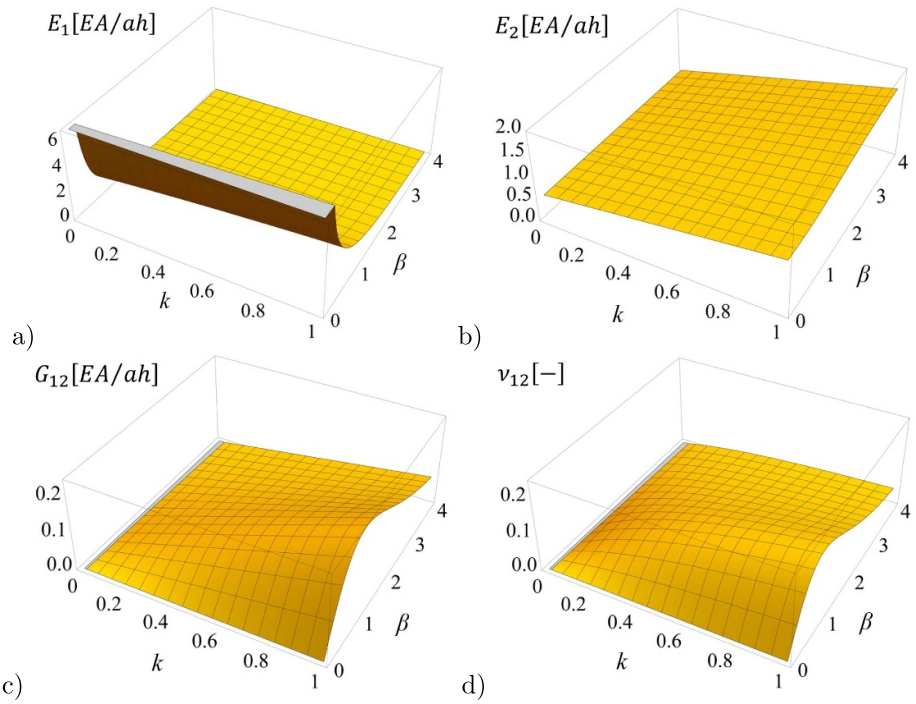


Figure 4. Graphical comparison of technical coefficients for Hexa-1: (a) Young's modulus E_1; (b) Young's modulus E_2; (c) shear modulus G_{12}; (d) Poisson's ratio ν_{12}.

Table 1. Values of the equivalent technical coefficients for six variants of the hexagonal module, calculated for $a = 7.5$ mm and $\sigma = 0.01$.

	Regular-1a	Regular-1b	Regular-1c	Rotated	Lower-1	Lower-2
	$d_c = 0.75$ $d_s = 1.5$ $\beta = 1$	$d_c = 0.75$ $d_s = 2.0$ $\beta = 1$	$d_c = 0.75$ $d_s = 2.5$ $\beta = 1$	$d_c = 0.75$ $d_s = 1.5$ $\beta = 1$	$d_c = 0.75$ $d_s = 1.5$ $\beta = 0.75$	$d_c = 0.75$ $d_s = 1.5$ $\beta = 0.5$
E_1 [$^*E$]	0.182 415	0.228 849	0.277 639	0.035 511	0.249 775	0.383 784
E_2 [$^*E$]	0.116 230	0.140 424	0.167 037	0.037 595	0.114 905	0.114 007
G_{12} [$^*E$]	0.008 109	0.005 853	0.004 447	0.023 830	0.007 256	0.005 563
ν_{12}	0.069 556	0.041 637	0.026 610	0.735 729	0.063 032	0.048 759
ν_{21}	0.044 319	0.025 549	0.016 010	0.778 901	0.028 997	0.014 485
$\kappa 11$	0	0	0	0.785 911	0	0
$\kappa 21$	0	0	0	−0.933 662	0	0
$\lambda 11$	0	0	0	0.527 388	0	0
$\lambda 12$	0	0	0	−0.591 810	0	0
I_1 [$^*E$]	0.315 787	0.381 373	0.453 760	0.315 787	0.379 860	0.509 269
I_2 [$^*E^2$]	0.026 126	0.036 498	0.050 352	0.026 126	0.034 055	0.049 327
I_3 [$^*E^3$]	0.000 345	0.000 377	0.000 413	0.000 345	0.000 417	0.000 487

assess susceptibility to other load cases (horizontal and shear).

The results presented in table 1 allow us to draw the following remarks:

Regular-1a

- $E_2 < E_1$: greater deformations under vertical compression (y direction) than under horizontal compression (x direction),
- small G_{12}: large deformations under shear,
- small ν_{12} and ν_{21}: very small extensions in transverse directions under compression,
- zero κ_{11} and κ_{21}: no shear deformations under compression,
- zero λ_{11} and λ_{12}: no extension under shear.

Regular-1b,c (compared to Regular-1a)

- higher E_1 and E_2: smaller deformations under compression, by 20% (Regular-1b) and 34%

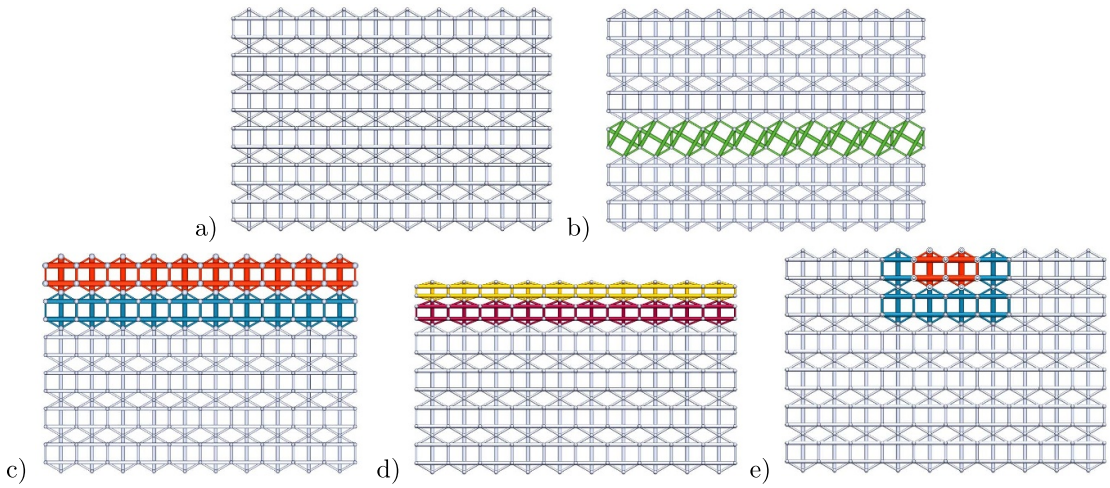


Figure 5. Hexagonal lattices: (a) M.hex.1.uni1; (b) M.hex.1.gra1; (c) M.hex.1.gra2; (d) M.hex.1.gra4; (e) M.hex.1.gra5.

(Regular-1c) in x direction, and by 17% (Regular-1b) and 30% (Regular-1c) in y direction,
- lower G_{12}: greater deformations under shear, by 39% (Regular-1b) and 82% (Regular-1c),
- lower ν_{12} and ν_{21}: even smaller extensions in transverse directions under compression,
- zero κ_{11} and κ_{21}: no shear deformations under compression,
- zero λ_{11} and λ_{12}: no extension under shear.

Rotated (compared to Regular-1a)

- lower E_1 and E_2: greater deformations under compression, five times higher in x direction and three times in y direction,
- higher G_{12}: significantly lower deformations under shear, by 66%,
- very high ν_{12} and ν_{21}: significant extensions in transverse directions under compression,
- non-zero κ_{11} and κ_{21}: shear deformations under compression,
- non-zero λ_{11} and λ_{12}: extensions under shear.

Lower-1,2 (compared to Regular-1a)

- higher E_1 and similar E_2: smaller deformations under compression in x direction, by 27% (Lower-1) and 52% (Lower-2), very similar deformations in y direction (difference less than 2%),
- lower G_{12}: greater deformations under shear, by 12% (Lower-1) and 46% (Lower-2),
- small ν_{12} and ν_{21}: very small transverse deformations under compression,
- zero κ_{11} and κ_{21}: no shear deformations under compression,
- zero λ_{11} and λ_{12}: no extension under shear.

Using the hexagonal modules described above, five different configurations of 2D lattices were proposed for further experimental testing:

- M.hex.1.uni1 (figure 5(a)) – regular lattice constructed from Regular-1a cells,
- M.hex.1.gra1 (figure 5(b)) – graded lattice with one layer of Rotated cells,
- M.hex.1.gra2 (figure 5(c)) – graded lattice with uppermost layer of Regular-1c cells and second (from top) layer of Regular-1b cells,
- M.hex.1.gra4 (figure 5(d)) – graded lattice with uppermost layer of Lower-2 cells and second (from top) layer of Lower-1 cells,
- M.hex.1.gra5 (figure 5(e)) – graded lattice with centric gradation using Regular-1c cells (two middle cells of the uppermost layer) and Regular-1b cells (six surrounding cells in top two layers).

External loads are to be applied to the top edge of the lattices, while free-standing boundary conditions are imposed on the bottom edge. The details are described in the next section.

5. AM and experimental setup

Tensegrity-inspired lattices based on hexagonal units were additively manufactured using the ProJet MJP 2500 Plus printer by 3D Systems. MultiJet Printing (MJP) is a 3D printing method based on inkjet technology that employs piezoelectric printheads to deposit layers of UV-curable resin and wax as a support material. This technology is well suited for fabricating tensegrity specimens due to the ease of post-processing: supports are removed by melting the wax, and even intricate features or complex architectures can be cleaned effectively without causing damage.

The post-processing steps in the MJP technology include wax removal and cleaning and drying of the specimens. After printing, the specimens undergo a wax removal phase using the MJP EasyClean system. This system employs steam and soy-based oil to gently melt the wax supports without applying mechanical force, preserving the integrity of intricate

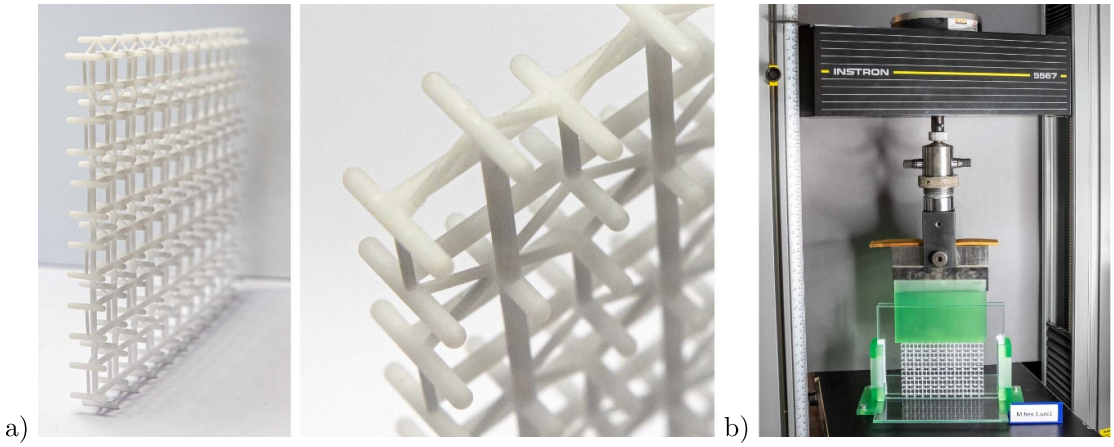


Figure 6. Hexagonal lattice: (a) 3D printed specimen: side view and detail, (b) test setup.

features and reducing the risk of damage. Following removal of the bulk support material, the parts are subjected to a fine cleaning process. This involves the use of solvents to remove residual wax traces from the surface areas. Subsequently, the specimens are dried to remove any remaining solvents. The post-processing steps are designed to minimise any negative impact on the mechanical properties of the printed lattices. By avoiding high-pressure or abrasive cleaning methods, structural integrity and surface quality are maintained. However, it is important to note that any residual support material or incomplete cleaning could potentially affect the mechanical performance of the specimens. Therefore, special attention must be paid to the post-processing procedures to ensure the accuracy and reliability of the experimental results.

The specimens tested in this study were manufactured from VisiJet M2R-WT resin, which according to the technical data sheet has the following properties: tensile modulus of 2000 MPa, tensile strength of 39 MPa, and extension at break of 20% (values determined according to ASTM D638 standard [42]). Figure 6(a) shows one of the manufactured lattices. It can be seen that the cables and struts lie in two parallel vertical planes, positioned as close to each other as possible without making contact between the struts. The transverse bars visible in the photographs are auxiliary elements that stabilise the 2D lattice in the transverse direction, ensuring that it undergoes only in-plane deformations under compression and helping to minimise friction via point contacts at the nodes; they do not serve a structural purpose.

All manufactured lattices were subjected to uniaxial compression using a universal testing machine INSTRON 5567 equipped with a load cell INSTRON 2525-816 that provides a measurement accuracy of 0.5 N. The experimental setup is presented in figure 6(b). The tests were carried out under displacement control at a constant rate of 5 mm/min, with the corresponding force recorded directly by the testing machine. The specimens were placed between two grease-lubricated glass plates to ensure in-plane compression, while minimising friction and allowing free movement of the nodes, so that the soft modes of tensegrity were not blocked by the boundary conditions.

6. Experimental investigation of selected lattices

In this section, the results of the experimental tests performed on additively manufactured lattices are discussed. Each example presented below refers to a different configuration of the lattice (figure 5) or a different load type applied: uniformly distributed (Examples 1–4) or concentrated (Examples 5–7).

For each analysed case, a force–displacement graph is presented, showing the mean curve together with the standard deviation, obtained from compression tests on five specimens of the given lattice. The force in the graphs is the resultant force applied by the load cell of the testing machine, and the displacement is the displacement measured along the vertical axis of the load cell. In addition, the deformations of the lattices were analysed. The photographs of the deformed lattices presented in this section were taken after the machine had been stopped, so they correspond to the maximum vertical displacement registered in each case. The complete raw experimental data corresponding to all lattice configurations and loading scenarios discussed in this section are openly available in the data repository [43].

It should be highlighted that AM technology produces lattices with rigid or semi-rigid nodes, unlike the continuum model that describes a pin-jointed cable-strut structure. This discrepancy may lead to differences in load transfer and deformation mechanisms, as rigid nodes increase the overall stiffness and constrain local rotations compared to the idealised pin-jointed model. However, several studies have

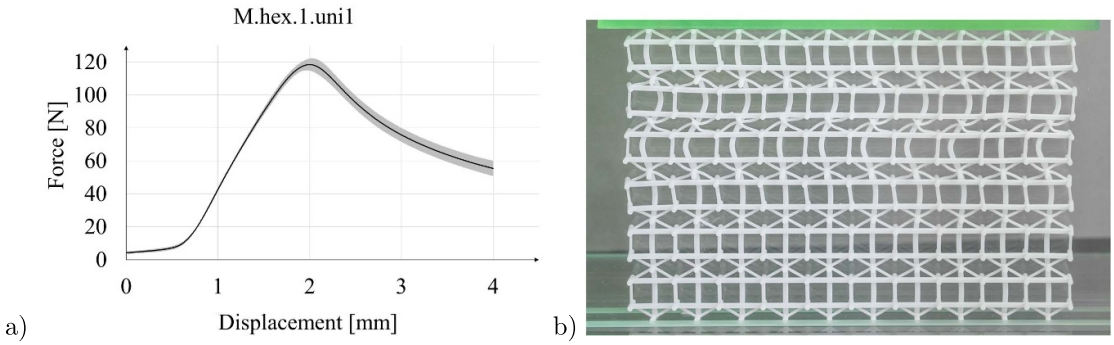


Figure 7. Regular lattice (M.hex.1.uni1) under a uniformly distributed load: (a) force–displacement graph; (b) deformation during the test.

shown that rigid node lattices can still reproduce the characteristic mechanical response of pure tensegrity systems [26–28]. Therefore, the continuum model remains a useful tool for estimating mechanical properties and guiding the design of graded lattices, even if quantitative agreement with the experiments is only approximate.

6.1. Example 1

Example 1 concerns a regular lattice without gradations in mechanical properties (see figure 5(a), tested under a uniformly distributed load applied to the entire upper edge of the structure. The results of the compression tests are presented in figure 7.

A fairly large initial portion of the force–displacement graph observed in figure 7(a) corresponds to the seating of the specimens. The lattice reaches the load-bearing capacity of 118.51 N at a displacement of 1.99 mm. The stiffness of the lattice, determined based on the slope of the linear portion of the graph, is 96.392 N mm^{-1}.

Figure 7(b) shows the deformation of the lattice under uniform load. The deformation of layers no. 1 and 4, counting from the top, is smaller than that of layers no. 2 and 3, which were damaged in the test. Layers no. 5 and 6 constitute an area of mechanical stability and are essentially not deformed. The deformation of the structure involves very small shear deformations and negligible horizontal extensions, which is reflected in the substitute elasticity coefficients: zero κ coefficients and small Poisson's ratios of the Regular-1a module (see table 1).

6.2. Example 2

Example 2 concerns a graded lattice with an intermediate layer constructed from rotated modules (see figure 5(b), tested under a uniformly distributed load applied to the entire upper edge of the structure. The results of the compression tests are presented in figure 8.

The initial portion of the force–displacement graph (figure 8(a)) corresponding to the seating of the specimens is much smaller than in the regular lattice. The lattice reaches the load-bearing capacity of 48.57 N (41% of the value obtained for the regular lattice) at a displacement of 2.41 mm. The stiffness of the lattice is 30.995 N mm^{-1}, which is 32% of the value for the regular lattice. The decrease in stiffness and load-bearing capacity of the structure compared to the regular lattice is fully justified by the value of Young's modulus E_2 of the Rotated module (see table 1), which is 32% of the value for the Regular-1a module. It is worth noting that the values of the invariants of the elasticity tensor are the same in the cells of the rotated and non-rotated layers.

The application of the rotated layer (no. 4 counting from the top) leads to the deformation of only that layer in the compression test. Having significantly lower Young's modulus E_2 than non-rotated layers, layer no. 4 is soft in the vertical direction, so the principle that the weakest element (a layer in this case) determines the damage applies. In addition, the Rotated module has an order of magnitude higher Poisson's ratios compared to the Regular-1a module and non-zero κ coefficients (see table 1), which is visible in the deformed lattice in the form of significant transverse deformations of the soft layer. The area of mechanical stability covers all layers above and below layer no. 4.

6.3. Example 3

Example 3 concerns a graded lattice with gradation of two upper layers via the strut thickness (see figure 5(c), tested under a uniformly distributed load applied to the entire upper edge of the structure. The results of the compression tests are presented in figure 9.

The initial portion of the force–displacement graph (figure 9(a)) corresponding to the seating of the specimens is very small. The lattice reaches the load-bearing capacity of 187.17 N (increase by 58% compared to the regular lattice) at a displacement of 1.31 mm. A significant increase in stiffness compared to the regular lattice is observed: initially 179.010 N mm^{-1} (increase by 86%) and from the middle 159.030 N mm^{-1} (increase by 64%). The

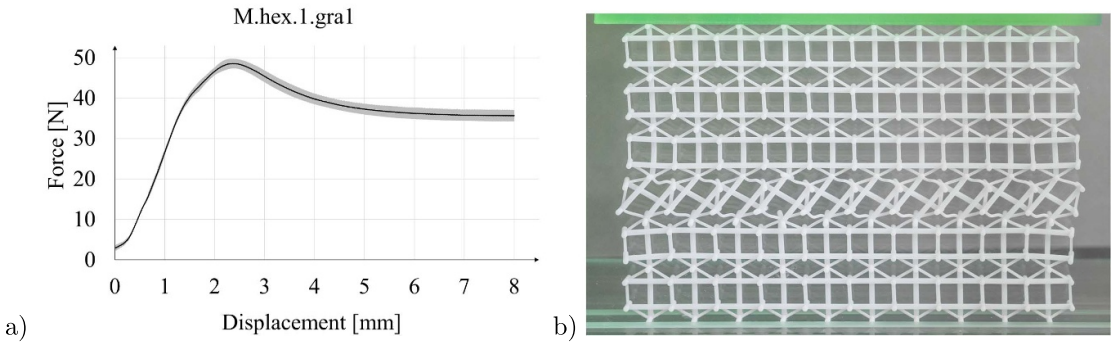


Figure 8. Graded lattice (M.hex.1.gra1) under a uniformly distributed load: (a) force–displacement graph; (b) deformation during the test.

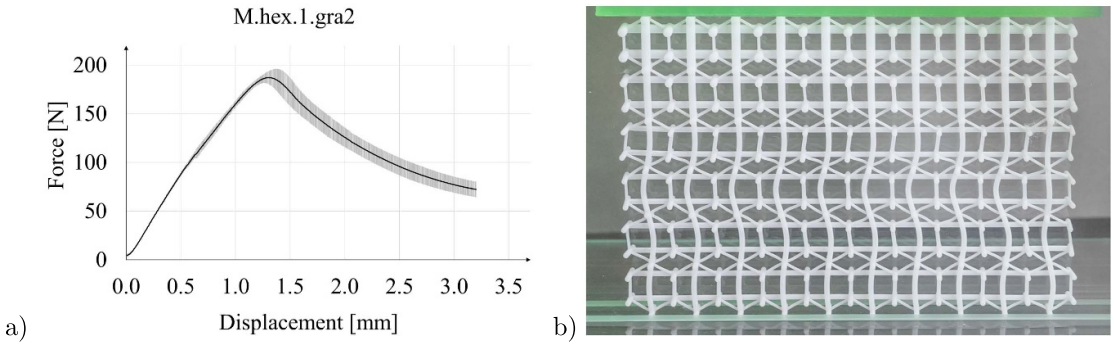


Figure 9. Graded lattice (M.hex.1.gra2) under a uniformly distributed load: (a) force–displacement graph; (b) deformation during the test.

increase in stiffness and load-bearing capacity of the lattice is justified by the values of Young's modulus E_2 of the Regular-1b and Regular-1c modules (see table 1), which are by 21% (Regular-1b) and 44% (Regular-1c) higher than the value for the Regular-1a module. Similar indicators can be observed in the first invariant of the elasticity tensor, which is related to the sum of Young's and shear moduli. A greater increase can be observed in the second invariant of the elasticity tensor, which is responsible for the sum of mutual couplings of the compression in each direction and shear.

The reinforcement of the first two layers caused the deformation to shift downward and changed the location of the area of mechanical stability (figure 9(b)). The deformation of layers no. 3 and 6 is smaller than that of layers no. 4 and 5, which are damaged. Layers no. 1 and 2 constitute the area of mechanical stability. The lattice still exhibits very small shear deformations and negligible horizontal extensions, since the substitute elasticity coefficients of the reinforced layers are even smaller than those of the regular modules (see table 1).

6.4. Example 4

Example 4 concerns a graded lattice with gradation of two upper layers via the module height (see figure 5(d), tested under a uniformly distributed load applied to the entire upper edge of the structure. The results of the compression tests are presented in figure 10.

The initial portion of the force–displacement graph (figure 10(a)) corresponding to the seating of the specimens is very small. The lattice reaches the load-bearing capacity of 138.46 N (increase by 17% compared to the regular lattice) at a displacement of 1.28 mm. The stiffness of the lattice is 124.320 N mm^{-1}, which is by 29% higher than that of the regular lattice. The increase in stiffness and load-bearing capacity of the structure compared to the regular lattice is partly justified by the increase in Young's modulus in the horizontal direction in the reinforced layers with an almost unchanged modulus in the vertical direction. There is also a visible increase in all invariants of the elasticity tensor.

The reinforcement of the first two layers caused the deformation to shift downward and changed the location of the area of mechanical stability (figure 10(b)). The deformation of layer no. 1 is smaller than that of layer no. 2 and much smaller than that of layers no. 3–6, which are damaged. Only layer no. 1 constitutes the area of mechanical stability. The lattice still exhibits very small shear deformations and negligible horizontal extensions, since the substitute elasticity coefficients of the reinforced layers are smaller than those of the regular modules (see table 1).

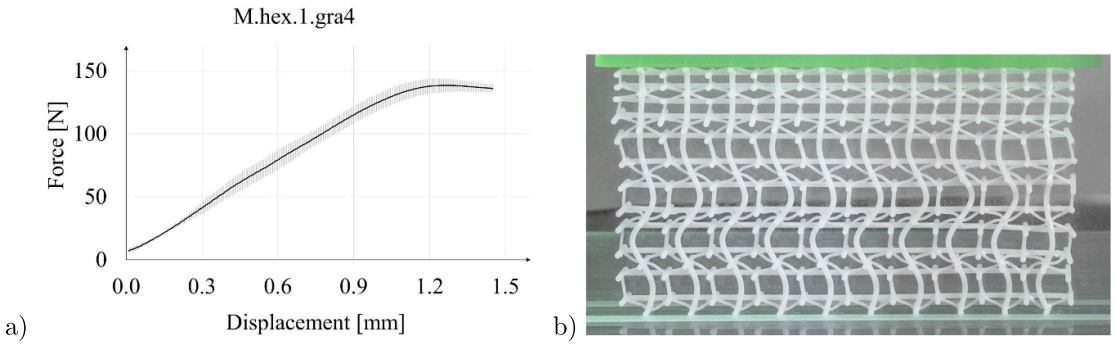


Figure 10. Graded lattice (M.hex.1.gra4) under a uniformly distributed load: (a) force–displacement graph; (b) deformation during the test.

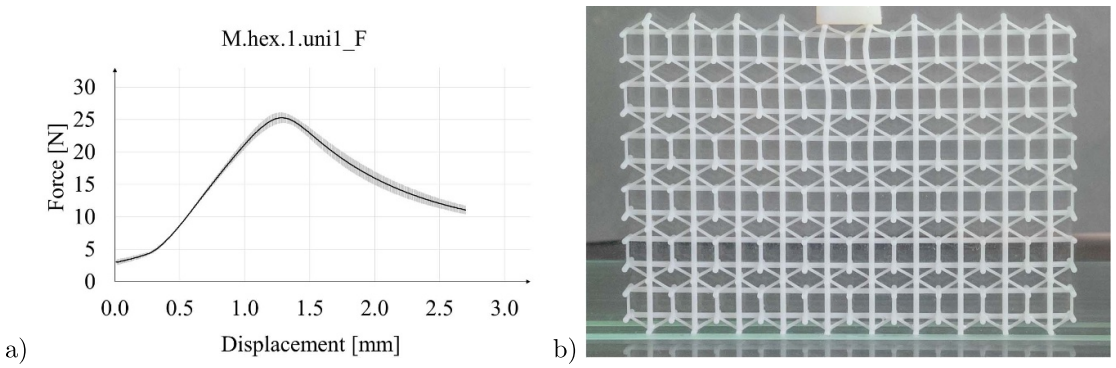


Figure 11. Regular lattice (M.hex.1.uni1) under a concentrated load: (a) force–displacement graph; (b) deformation during the test.

6.5. Example 5

Example 5 concerns a regular lattice without gradations in mechanical properties (see figure 5(a)), tested under a concentrated load applied to two middle upper nodes of the structure. The results of the compression tests are presented in figure 11.

A fairly large initial portion of the graph corresponding to the seating of the specimens is observed in figure 11(a). The lattice reaches the load-bearing capacity of 25.29 N at a displacement of 1.29 mm. The stiffness of the lattice is 24.870 N mm^{-1}. Under concentrated loads, the load-bearing capacity is almost five times lower than under a uniformly distributed load, and the stiffness of the lattice is four times lower. The loss of load-bearing capacity occurs at significantly lower displacements.

The deformation of the lattice (figure 11(b)) is local and affects two layers of cells adjacent to the load (four modules). The area of mechanical stability covers the remaining modules. Two cells under load are damaged. Apart from locally deformed cells, no shear deformations or cell extensions are observed, which is consistent with low elasticity coefficients obtained in the continuum model (see table 1).

6.6. Example 6

Example 6 concerns a graded lattice with gradation of two upper layers via the strut thickness (see figure 5(c), tested under a concentrated load applied to two middle upper nodes of the structure. The results of the compression tests are presented in figure 12.

The initial portion of the force–displacement graph (figure 12(a)) corresponding to the seating of the specimens is very small. A local extremum in the force–displacement graph of 61.66 N (2.44 times the load-bearing capacity of the regular lattice) is achieved at a displacement of 1.18 mm, followed by a non-linear amplification to reach the final load-bearing capacity of 72.78 N (2.88 times that of the regular lattice) at a displacement of 2.25 mm. The stiffness of the lattice is 55.815 N mm^{-1}, which is 2.24 times higher than the value for the regular lattice. The increase in stiffness and load-bearing capacity of the structure is not justified by the substitute values of the elasticity coefficients. It can be justified with an increase in the number of deformed cells by nearly 2.5 times—the damage is no longer local under the applied load.

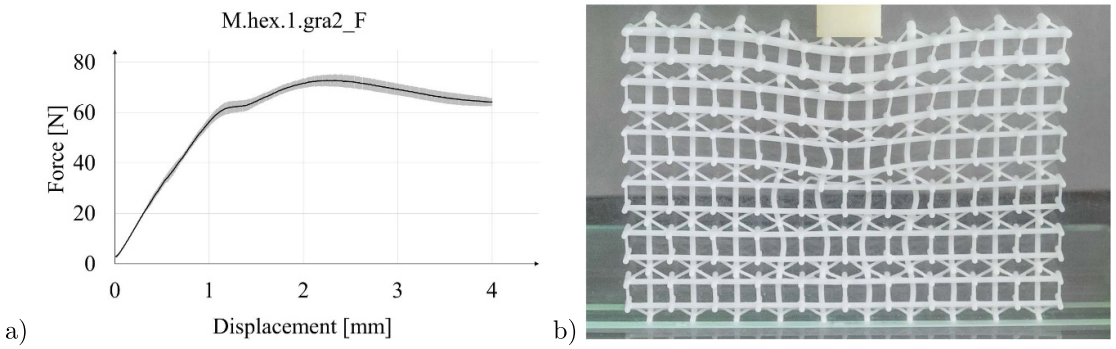


Figure 12. Graded lattice (M.hex.1.gra2) under a concentrated load: (a) force–displacement graph; (b) deformation during the test.

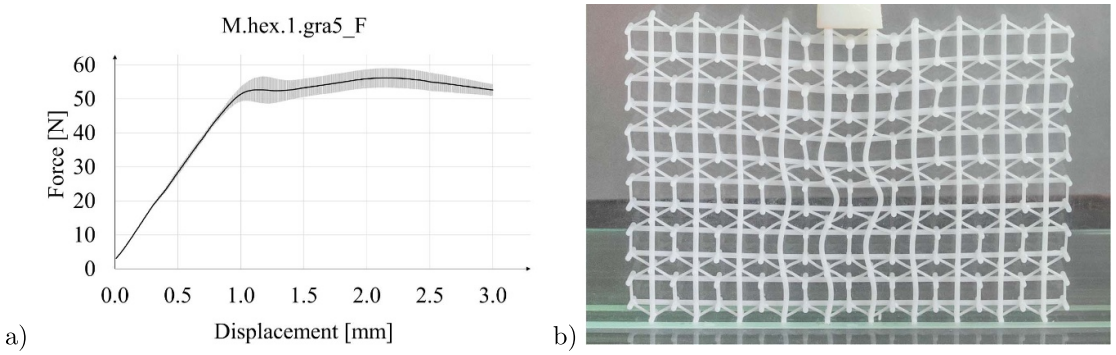


Figure 13. Graded lattice (M.hex.1.gra5) under a concentrated load: (a) force–displacement graph; (c) deformation during the test.

A significant increase in stiffness and load-bearing capacity is achieved at the expense of an increased deformation area (figure 12(b)). The uniform reinforcement of the first two layers causes deformation that affects the entire central part of the structure. The reinforced layers deform gently and damage occurs in four central columns in layers no. 3–5. Layer no. 6 undergoes only slight deformation in the centre. The area of mechanical stability is therefore smaller than in the regular lattice. The structure still shows significant local shear deformations and negligible horizontal extensions, which results from the concentrated nature of the load.

6.7. Example 7

Example 7 concerns a graded lattice with centric gradation of two upper layers via the strut thickness (see figure 5(e), tested under a concentrated load applied to two middle upper nodes of the structure. The results of the compression tests are presented in figure 13.

The initial portion of the force–displacement graph (figure 13(a)) corresponding to the seating of the specimens is still very small. A local extremum in the force–displacement graph of 52.69 N (2.08 times the load-bearing capacity of the regular lattice) is achieved at a displacement of 1.16 mm, followed by a non-linear amplification to reach the final load-bearing capacity of 56.18 N (2.22 times that of the regular lattice) at a displacement of 2.19 mm. The stiffness of the lattice is 51.274 N mm^{-1}, which is 2.06 times higher than the value for the regular lattice. The results obtained with the local reinforcement of 8 cells are only slightly lower than those obtained with the reinforcement of 20 cells in two layers (Example 6).

Centric reinforcement of two layers near the load causes deformation similar to that obtained in Example 6, covering four middle columns with the damage occurring in Layer no. 4 (figure 13(b)). Layer no. 6 undergoes greater deformation than in Example 6. The area of mechanical stability includes, as before, the first three and last three columns of the lattice. As in the previous example, it is difficult to quantitatively justify the results using the substitute elasticity coefficients. However, it is clear that reinforcing entire layers is not the right approach. Almost the same effect can be achieved by locally reinforcing the area near the load.

6.8. Failure modes, load-bearing capacity, and stiffness

In the seven examples presented above, several failure modes can be identified, and their locations within the lattices can be indicated. Under vertical loading, cells fail due to vertical compression and shear deformation, and in several cases these

Table 2. Classification of failure modes in cells.

Mode	Symbol	Colour
Compression strong	CS	Deep yellow
Compression medium	CM	Light yellow
Shear strong	SS	Deep blue
Shear medium	SM	Light blue
Mixed	MM	Green
Mechanically stable	ST	Light grey

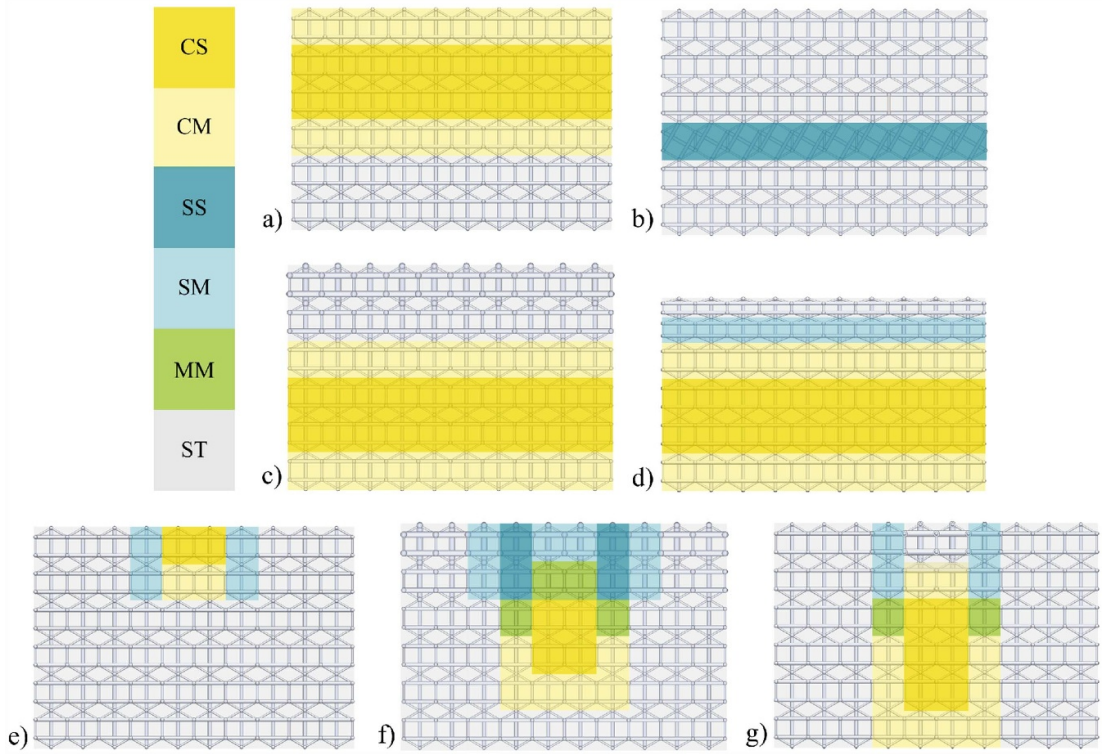


Figure 14. Failure modes: (a) regular lattice (M.hex.1.uni1) – uniform load; (b) graded lattice (M.hex.1.gra1) – uniform load; (c) graded lattice (M.hex.1.gra2) – uniform load; (d) graded lattice (M.hex.1.gra4) – uniform load; (e) regular lattice (M.hex.1.uni1) – concentrated load; (f) graded lattice (M.hex.1.gra2) – concentrated load; (g) graded lattice (M.hex.1.gra5) – concentrated load.

failure modes occur simultaneously. Cells that do not deform (i.e. show no signs of failure) define mechanically stable regions. For a concise presentation and schematic illustration of the failure modes observed during testing, six characteristic failure types are proposed and summarised in table 2.

Figure 14 shows a graphical interpretation of the failure modes identified in the tested lattices. Strong and medium compression failure modes dominate in the regular lattice under uniform load (figure 14(a)). A similar failure pattern, shifted toward the lower layers, is observed in the graded lattices (figures 14(c) and (d)). For the second gradation, additionally, slight shear can be recognised in the second layer (figure 14(d)). In contrast, in the lattice with a soft layer, a pure strong shear failure mode is observed (figure 14(b)).

Much more diverse failure modes are observed in the lattices tested under concentrated loads (figures 14(e)–(g)). Cells in the area below the applied force fail due to compression, shear, or both, depending on the case.

A medium (from 119 N to 187 N, for M.hex.1.gra2) or slight (from 119 N to 138 N, for M.hex.1.gra4) increase in the load-bearing capacity for graded lattices under uniform load can be explained by the increased stiffness of the two upper layers, without a significant change in the failure modes shifted to the lower layers. The slight shear in the second layer observed in figure 14(d) may explain the smaller increase in the load-bearing capacity for the second type of gradation (M.hex.1.gra4).

The significant increase in the load-bearing capacity for graded lattices under concentrated loads can be explained by the delocalisation of deformations. In the uniform lattice (figure 14(e)), strong deformation is observed in two cells and medium deformation in six cells located near the applied load. In contrast, for graded lattices (figures 14(f) and (g)), the load is transferred to the lower layers, resulting in

Table 3. Summary of the load-bearing capacity and stiffness of the lattices.

Example	Lattice type	Load type	Stiffness of the lattice [N mm^{-1}]	Load-bearing capacity [N]
1	M.hex.1.uni1	uniform	96.392	118.51
2	M.hex.1.gra1	uniform	30.995	48.57
3	M.hex.1.gra2	uniform	179.010/159.030	187.17
4	M.hex.1.gra4	uniform	124.320	138.46
5	M.hex.1.uni1	concentrated	24.870	25.29
6	M.hex.1.gra2	concentrated	55.815	72.78
7	M.hex.1.gra5	concentrated	51.274	56.18

six to eight strongly deformed cells and sixteen cells with medium deformation. Consequently, the load-bearing capacity increased from 25 N to 73 N (four cells with strong failure) for the layered gradation (M.hex.1.gra2) and to 56 N (six cells with strong failure) for the radial gradation (M.hex.1.gra5).

The load-bearing capacity and stiffness of all the lattices tested experimentally are summarised in table 3. The stiffness was determined based on the slope of the linear portion of the graphs presented in figures 7–13, and therefore, is expressed in [N mm^{-1}].

7. Conclusions

The present study focused on modelling 2D tensegrity-inspired lattices with graded mechanical properties, with the objective of enabling control over stiffness, load-bearing capacity, and the extent of mechanically stable regions, understood as areas that remain practically undeformed. The computational framework developed in this work is intended as a tool for the preliminary evaluation of the mechanical behaviour of additively manufactured systems and for formulating recommendations on the arrangement of cells and layers in 2D lattices. As the proposed computational models were not designed to simulate real metamaterials, calibration and validation against experimental results were not performed. Instead, preliminary experimental investigations on additively manufactured tensegrity-inspired lattices in a medium scale were carried out to assess their structural response and guide the modelling approach. Five regular and graded lattices, each composed of a 10×6 array of hexagonal modules, were tested under both uniform and concentrated compression loading, and the effects of gradation on their mechanical behaviour were examined in detail.

Following the results developed in this paper, several conclusions can be drawn. For 2D hexagonal lattices, it is possible to derive closed-form formulas for the equivalent mechanical properties within a continuum model. These formulas are valid for single cells or for layers composed of identical cells, and they provide a useful basis for modelling lattices with graded mechanical properties. The description of the soft and extremal mechanical properties obtained for the rotated Hexa-4 module is particularly valuable in this regard.

It was demonstrated that graded tensegrity-inspired lattices can be fabricated using AM technology and subsequently tested in a universal testing machine. The MJP technology was found to be best suited for fabricating tensegrity specimens due to the ease of post-processing, as the wax supports are simply melted and complex features can be cleaned effectively without damage.

By modifying the mechanical properties of lattices, it is possible to control their load-bearing capacity and stiffness, as well as to create areas of mechanical stability. Modelling the response of graded lattices under compression with uniformly distributed loads is more straightforward and predictable. It reflects the principle of failure of the weakest element of the lattice and allows for the intentional modification of mechanically stable regions.

In contrast, the action of concentrated loads presents greater complexity, but also offers deeper cognitive insights. Under such loading conditions, localised deformations were observed at the point of application of the load, together with the emergence of significant regions of mechanical stability.

The extremal properties of tensegrity-inspired cells highlight the strong influence of the module geometry and the cable-to-strut stiffness ratio k on the results. Further investigations are therefore needed to analyse the role of self-stress in additively manufactured lattices and its effect on their mechanical behaviour.

The results at a medium scale are promising and motivate further research into the modelling of the static and dynamic properties of graded tensegrity-inspired lattices, with an aim of designing advanced additively manufactured metamaterials for various engineering applications. Future work should include dynamic and fatigue testing to better understand the long-term and rate-dependent behaviour of these systems. Moreover, extending the computational framework to incorporate application-oriented designs and to account for material nonlinearity and dynamic effects could further improve its practical relevance.

It should be noted that the present study has some limitations. As discussed in section 6, the continuum model assumes idealised pin-jointed connections, whereas the printed lattices have rigid nodes.

In addition, the analysis was performed at a medium scale, chosen to accommodate six layers of cells, allow visual observation of deformations, and comply with the constraints of the AM process. The effects of scale on the mechanical response of metamaterials will need to be carefully analysed in future studies when moving toward the material scale.

Funding

This research was funded in whole by National Science Centre, Poland, Grant No. 2022/47/D/ST8/00466. For the purpose of Open Access, the author has applied a CC-BY public copyright licence to any Author Accepted Manuscript (AAM) version arising from this submission.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.17632/xhsr5hb8v9.1; https://doi.org/10.17632/d8ymhhj677.1.

Author contributions

Wojciech Gilewski 0000-0001-6374-8284
Conceptualization (equal), Data curation (equal), Formal analysis (equal), Investigation (equal), Methodology (equal), Software (equal), Supervision (equal), Validation (equal), Writing – original draft (equal)

Adam Zawadzki 0000-0003-0491-8831
Conceptualization (equal), Data curation (equal), Investigation (equal), Methodology (equal), Software (equal), Validation (equal), Visualization (equal), Writing – review & editing (equal)

Marta Thammavong
Data curation (equal), Investigation (equal), Software (equal), Visualization (equal)

Anna Al Sabouni-Zawadzka 0000-0003-2688-8442
Conceptualization (equal), Funding acquisition (equal), Investigation (equal), Methodology (equal), Project administration (equal), Resources (equal), Software (equal), Supervision (equal), Writing – review & editing (equal)

References

[1] Surjadi J U, Gao L, Du H, Li X, Xiong X and Fang N X *et al* 2019 Mechanical metamaterials and their engineering applications *Adv. Eng. Mater.* **21** 1800864
[2] Lee T U, Chen Y, Heitzmann M T and Gattas J M 2021 Compliant curved-crease origami-inspired metamaterials with a programmable force-displacement response *Mater. Des.* **207** 109859
[3] W W S Ma *et al* 2025 Multi-physical lattice metamaterials enabled by additive manufacturing: design principles, interaction mechanisms and multifunctional applications *Adv. Sci.* **12** 2405835
[4] Akinwande D *et al* 2017 A review on mechanics and mechanical properties of 2D materials—Graphene and beyond *Extreme Mech. Lett.* **13** 42–77
[5] Androulidakis C, Zhang K, Robertson M and Tawfick S 2018 Tailoring the mechanical properties of 2D materials and heterostructures *2D Mater.* **5** 032005
[6] Kim J H, Jeong J H, Kim N, Joshi R and Lee G H 2018 Mechanical properties of two-dimensional materials and their applications *J. Phys. D: Appl. Phys.* **52** 083001
[7] Backes C *et al* 2020 Production and processing of graphene and related materials *2D Mater.* **7** 022001
[8] Chen M, Christmann A, Muniz A, Ramasubramaniam A and Maroudas D 2020 Molecular-dynamics analysis of nanoindentation of graphene nanomeshes: implications for 2D mechanical metamaterials *ACS Appl. Nano Mater.* **3** 3613–24
[9] Wu W, Hu W, Qian G, Liao H, Xu X and Berto F 2019 Mechanical design and multifunctional applications of chiral mechanical metamaterials: a review *Mater. Des.* **180** 107950
[10] Ni X, Guo X, Li J, Huang Y, Zhang Y and Rogers J 2019 2D Mechanical metamaterials with widely tunable unusual modes of thermal expansion *Adv. Mater.* **10** e1905405
[11] Jiao W, Shu H, TOURNAT V, Yasuda H and Raney J 2024 Phase transitions in 2D multistable mechanical metamaterials via collisions of soliton-like pulses *Nat. Commun.* **15** 15
[12] Choi C, Burggraf J, Tyedmers A, Gonder S and Yu B 2024 The role of topological defects in the mechanics of additive manufactured 2D architected honeycomb metamaterials *Proc. 63rd Conf. of Metallurgists*
[13] Li X, Gao L, Zhou W and Wang Y 2019 Novel 2D metamaterials with negative Poisson's ratio and negative thermal expansion *Extreme Mech. Lett.* **30** 100498
[14] Wang H, Zhang Y, Lin W and Qin Q H 2020 A novel two-dimensional mechanical metamaterial with negative Poisson's ratio *Comput. Mater. Sci.* **171** 109232
[15] Tian J, Tang K, Chen X and Wang X 2022 Machine learning-based prediction and inverse design of 2D metamaterial structures with tunable deformation-dependent Poisson's ratio *Nanoscale* **14** 14–40
[16] Mizzi L, Grasselli L, Spaggiari A, Gatt R, Farrugia P S and Grima J 2023 Design of isotropic 2D chiral metamaterials based on monohedral pentagonal tessellations *Thin-Walled Struct.* **187** 110739
[17] Skelton R and Oliveira M 2009 *Tensegrity Systems* (Springer)
[18] Micheletti A 2025 Basic Tensegrity Concepts and Calculation Methods *Tensegrity Systems: Basic Concepts, Mechanical Metamaterials, Biotensegrity*, ed F Fraternali and J J Rimoli (Springer) pp 1–44
[19] Dong Y, Yuan X, Wang X, Samy A, Ma S and Dong S 2025 Simultaneous topology, configuration and prestress optimization for lightweight design of modular tensegrity chain structures *Thin-Walled Struct.* **212** 113184
[20] Micheletti A, Ruscica G and Fraternali F 2019 On the compact wave dynamics of tensegrity beams in multiple dimensions *Nonlinear Dyn.* **98** 2737–53
[21] de Castro Motta J, Garanger K and Rimoli J J 2024 Propagation of compression solitary waves on tensegrity-like lattices made of truncated octahedrons *Int. J. Non-Linear Mech.* **162** 104716
[22] Al Sabouni-Zawadzka A, Micheletti A, Kołodziejczak M and Zawadzki A 2025 Design and fabrication of modular tensegrity-like lattices with auxetic properties *Mater. Des.* **258** 114513
[23] Rieffel J, Valero-Cuevas F and Lipson H 2009 Automated discovery and optimization of large irregular tensegrity structures *Comput. Struct.* **87** 368–79

[24] Amendola A, Hernández-Nava E, Goodall R, Todd I, Skelton R E and Fraternali F 2015 On the additive manufacturing, post-tensioning and testing of bi-material tensegrity structures *Compos. Struct.* **131** 66–71
[25] Liu K, Wu J, Paulino G and Qi H 2017 Programmable deployment of tensegrity structures by stimulus-responsive polymers *Sci. Rep.* **7** 3511
[26] Pajunen K, Johanns P, Pal R K, Rimoli J J and Daraio C 2019 Design and impact response of 3D-printable tensegrity-inspired structures *Mater. Des.* **182** 107966
[27] Intrigila C, Micheletti A, Nodargi N A, Artioli E and Bisegna P 2022 Fabrication and experimental characterisation of a bistable tensegrity-like unit for lattice metamaterials *Add. Manuf.* **57** 102946
[28] Al Sabouni-Zawadzka A, Gilewski W and Zawadzki A 2024 Experimental investigations on mechanical properties of 3D-printed tensegrity-inspired metamaterials based on 4-strut simplex module *Arch. Civil Eng.* **70** 343–57
[29] Vangelatos Z, Micheletti A, Grigoropoulos C P and Fraternali F 2020 Design and testing of bistable lattices with tensegrity architecture and nanoscale features fabricated by multiphoton lithography *Nanomaterials* **10** 652
[30] Pajunen K, Celli P and Daraio C 2021 Prestrain-induced bandgap tuning in 3D-printed tensegrity-inspired lattice structures *Extreme Mech. Lett.* **44** 101236
[31] Zeng H, Mu R, Huo K, Zhao H, Wang K and Wang A 2023 A novel 3D-printable tensegrity-inspired metamaterial enabling dynamic attenuation *Int. J. Mech. Mater. Des.* **19** 1–19
[32] Intrigila C, Micheletti A, Nodargi N A and Bisegna P 2023 Mechanical response of multistable tensegrity-like lattice chains *Additive Manuf.* **74** 103724
[33] Al Sabouni-Zawadzka A, Gilewski W, Nazifi Charandabi R and Zawadzki A 2024 Stability of tensegrity-inspired structures fabricated through additive manufacturing *Compos. Struct.* **345** 118377
[34] Obara P, Kłosowska J and Gilewski W 2019 Truth and myths about 2D tensegrity trusses *Appl. Sci.* **9** 179
[35] Gilewski W and Kasprzak A 2014 3D continuum models of tensegrity modules with the effect of self-stress *11th World Congress on Computational Mechanics, 5th European Conf. on Computational Mechanics* (*Barcelona*)
[36] Gilewski W and Al Sabouni-Zawadzka A 2020 Equivalent mechanical properties of tensegrity truss structures with self-stress included *Eur. J. Mech.-A/Solids* **83** 103998
[37] Green A E and Zerna W 1968 *Theoretical Elasticity* (Oxford University Press)
[38] Al Sabouni-Zawadzka A 2022 *High Performance Tensegrity-Inspired Metamaterials and Structures* (CRC Press)
[39] Milton G and Cherkaev A 1995 Which elasticity tensors are realizable? *J. Eng. Mater. Technol.* **17** 483–93
[40] Zadpoor A A 2016 Mechanical meta-materials *Mater. Horiz.* **3** 371–81
[41] Al Sabouni-Zawadzka A and Gilewski W 2025 Continuum analysis of 2D tensegrity modules *Mendeley Data* (https://doi.org/10.17632/xhsr5hb8v9.1)
[42] ASTM D 638-14: standard test method for tensile properties of plastics 2014 ASTM International pp 1–17
[43] Al Sabouni-Zawadzka A, Zawadzki A, Thammavong M and Gilewski W 2025 2D graded tensegrity-inspired metamaterials with hexagonal cells - experimental tests results *Mendeley Data* (https://doi.org/10.17632/d8ymhhj677.1)